Final Project: C Programming Application

CS 366: *Systems Programming*

Instructor: Collin Engstrom

Due: Last Week of Class

Directions: For your final project, you and your group will be responsible for implementing a "real-world" application. You will need to use the C programming language as we've been doing in class. You will also need to write a *makefile* to go along with your code. You will need to work in groups of two or three to finish your work. You will also give a final presentation and demo at the end of the semester. Here are some additional notes to help you:

1. You must choose a real-world project idea that will be of sufficient length for a term project. Implementations will vary, but your code *must* include the following:
    a. At least 200 lines of non-whitespace code C code,
    b. ANSI C compliance (i.e., compiles with the "-ansi"),
    c. Sufficient documentation to discern what your code is doing,
    d. Good programming practice (including spacing, variable naming, etc.) and
    e. At least two different functions, including `main()`.

2. All of your programming must be done in the C programming language in the Linux environment on your VMs. This is the platform I will use to compile and run your code, if necessary.

3. You will also need to include a *makefile*, along with your implementation. We'll go through how to do this in class.

4. In your final presentation, you must note why you chose the application you made. What individuals or companies would be your target audience?

5. Your code must compile and be bug-free on your assigned Linux VM to receive full credit. This includes being free of memory leaks, memory overflows, etc. In your presentation, you should note what process you used for debugging. Were there any bugs that gave you particular difficulty.

6. Please include your function prototypes in your slideshow as well. You should include a discussion on how these functions interact with one another.

7. Also, are there any ethical issues you can identify with your software? Is there a potential for abuse?

8. For your analysis, you may choose any real-world application. Some examples include:
    - A game (e.g., hangman, guess a number, card game etc.),
    - A management system (for a bank, business, library),
    - An invoicing system,
    - An industry application that involves embedded systems,
    - ….

9. You will prepare a final presentation using PowerPoint. This presentation will discuss all the points laid out in these specifications and anything else you deem important. I will collect your final slideshow as part of your grade.

10. You are expected to foster communication skills—both with group mates and also with the broader class. Unless otherwise specified, each team member ***must*** present during final presentation.

11. Note that your project must be developed de novo (i.e., from the ground up) together as a team. All group members must contribute to each aspect of the project, including the final presentation. (All group members ***must*** *be present—in person—and speak for the presentation*.) Additionally, you should ***not*** rip code from the Internet, although you are encouraged to use online tutorials to aid you in coding up your project.

12. You are expected to complete and submit all project files (including your source code file(s) and PowerPoint file) on D2L *before* the day of your presentation. I will provide more details, including the deadline, closer to the end of semester.
    a. Projects not submitted by the specified deadline may face a 25% deduction of *total points*.

13. Please let me know if you have any questions or need clarification on any of these points. Otherwise, good luck!